

will become longer as the number of memory nodes increases. However, given the same training time, the frequency of review will also decrease accordingly. On the other hand, too few memory nodes can also make the temporal property in the sequence not clear enough. The R56-R20 experiment shows the highest accuracy when $k = 6$.

Value of k	ACC(%)
3	71.76(+2.70)
6	71.81(+2.75)
9	71.78(+2.72)

Table 7: Ablation studies on the number of memory nodes on CIFAR-100. The student and teacher models are ResNet-56 and ResNet-20 respectively.

Extensions

Visualizations. We visualize the deep features of student network WRN-16-2 (distilled from WRN-40-2 on CIFAR-100). The t-SNE (Fig. 5) results show that representations of TSKD are more separable than general KD, which proves that student trained by TSKD benefits from the discriminability of features.

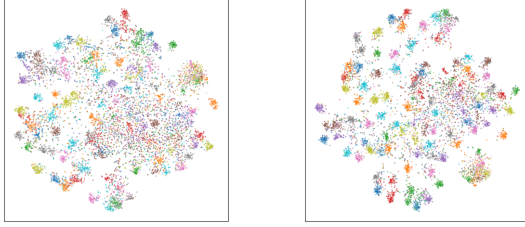


Figure 5: t-SNE of features learned by KD (left) and TSKD (right).

Efficiency. We compare the training cost of the proposed TSKD with state-of-the-art feature-based distillation methods. Specifically, we record the average training time per batch in a whole memory-review cycle. Different from existing methods which perform distillation throughout the entire training process, our TSKD, instead, only needs to distill in review nodes. Therefore, the computational costs of training student networks have been reduced. As the results reported in Fig. 5, our method has better trade-off between training time and accuracy.

Transferability. We evaluate the generalizability of distilled student network. A primary goal of representation learning is to acquire general knowledge that is not only useful in the current dataset, but also in dataset/tasks from other domains. Therefore, we test if the knowledge distilled by TSKD transfer well. In the experiment, we use WRN-16-2 as the frozen representation extractors, which is either trained from scratch on CIFAR-100 or distilled from WRN-40-2 with various KD

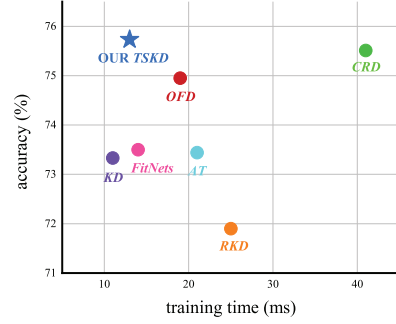


Figure 6: Training time (per batch) vs. accuracy on CIFAR-100. We set ResNet32x4 as the teacher and ResNet8x4 as the student.

methods. We then perform linear probing tasks on TinyImageNet². As the results reported in Table 8, TSKD outperforms other methods by a certain margin, demonstrating its strong generalizability.

baseline	KD	FitNets	CRD	ReviewKD	DKD	TSKD
33.7	33.9	33.5	35.6	36.6	37.1	38.5

Table 8: Comparison with previous methods on transferring representations from CIFAR-100 to TinyImageNet.

Conclusion

This paper deals with the problem of knowledge distillation from a novel perspective. We find that there exists temporal pattern in the evolution of network knowledge. Motivated by this observation, we propose Temporal Supervised Knowledge Distillation (TSKD) to solve two questions: (1) How to utilize the old knowledge to assist current learning? and (2) How to supervise the temporal learning process of network? The student trained by TSKD achieves significant improvements on CIFAR-100, ImageNet and MS-COCO datasets for image classification and object detection tasks. Besides, TSKD also shows superiority on training efficiency and knowledge transferability. We hope our work will help future research on knowledge distillation and interpretable deep learning.

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²<https://www.kaggle.com/c/tiny-imagenet>

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